Walmart Customer Analytics Dashboard using Power BI

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1. Introduction

This project focuses on analyzing Walmart customer data using Power BI to understand customer behavior, churn patterns, loyalty engagement, promotion effectiveness, and customer lifetime value. The objective is to takeout meaningful insights and provide business recommendations to improve customer retention and overall business performance.

2. Dataset Description

The analysis was performed using five datasets: Walmart Customer Transaction, Walmart Store Location, Walmart Loyalty Program, Walmart Customer Demographics, and Churn Labelled Customer.

3. Data Model & Relationships

Relationships were created using Customer_ID and Store_ID to connect all the tables.

4. Data Preparation & Calculations

Calculated Columns used:

Membership Duration Years = DATEDIFF(Customer_Demographics[Membership_Since],TODAY(),YEAR)

CLV = DIVIDE(Customer_Transactions[Total Amount Spend],RELATED(Customer_Demographics[Membership Duration Years]), 0)

CLV Segment = IF(Customer_Transactions[CLV] >= [Average CLV],"High-CLV","Low-CLV")

Days Since Last Purchase = DATEDIFF(Churn_Labelled_Customers[Last_Purchase_Date],TODAY(),DAY)

Customer Segment = SWITCH(TRUE(),'Customer_Transactions'[Purchase_Count] <= 3, "Low-Tier",'Customer_Transactions'[Purchase_Count] <= 8, "Mid-Tier", "High-Tier")

Age Group = SWITCH(TRUE(), Customer_Demographics[Age] < 25, "18-24",Customer_Demographics[Age] < 35, "25-34",Customer_Demographics[Age] < 50, "35-49","50+")

Measures used:

Churn Rate % = DIVIDE([Churned Customer],[Total Customer],0)

Churned Customers = CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag] = 1)

Repeat Customer = CALCULATE(DISTINCTCOUNT('Customer_Transactions'[Customer_ID]),Customer_Transactions[Purchase_Count] > 1)

Repeat Rate = DIVIDE([Repeat Customer],[Total Customer],0)

Retention Rate = DIVIDE([Retained Customer],[Total Customer])

Total Customers = DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID])

Avg Purchase Frequency = AVERAGE(Customer_Transactions[Purchase_Count])

Loyal Customer Purchase = CALCULATE(COUNT('Customer_Transactions'[Transaction_ID]),'Customer_Transactions'[Customer_Segment] = "High-Tier")

Average Purchase Amount = AVERAGE('Customer_Transactions'[Amount])

Promo % = DIVIDE([Promo Transaction],[Total Transaction],0)

Promo Transaction = CALCULATE(COUNT('Customer_Transactions'[Transaction_ID]),'Customer_Transactions'[Promotion_Applied] = "Yes")

Total Points Earned = SUM(Loyalty_Program[Points_Earned])

Total Points Redeemed = SUM(Loyalty_Program[Points_Redeemed])

Total Transaction = COUNT('Customer_Transactions'[Transaction_ID])

Avg Transaction Amount = AVERAGE(Store_Locations[Transactions_Amount])

Average CLV = AVERAGE(Customer_Transactions[CLV])
Customers were segmented into Low-Tier, Mid-Tier, and High-Tier based on purchase count.

Retained Customer = CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag] = 0)

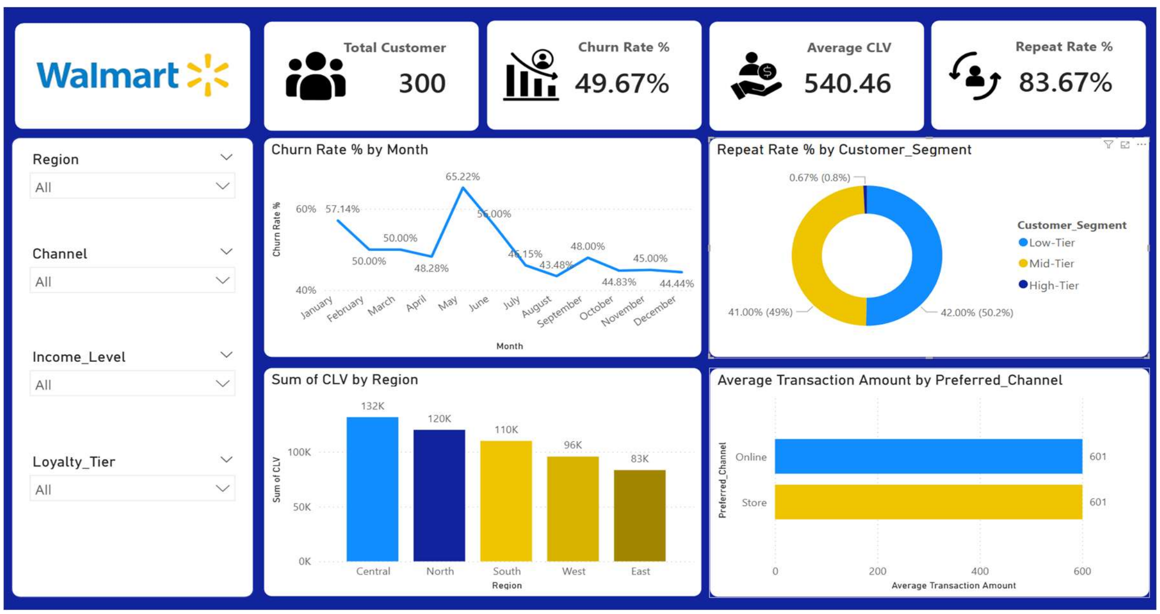
Retention Rate = DIVIDE([Retained Customer],[Total Customer])

I have also Created Slicers on Every Page Separately for better Experience.

5. Task 3 – KPI Overview

I have shown 4 major KPIs, which are Total Customer, Churn Rate, Average Customer Lifetime Value, and Repeat Rate.

Churn Rate tells us how many customers have stopped purchasing.
Repeat Rate tells us how many customers are returning and buying again.
And Average CLV tells us the long-term value generated by customers.



6. Task 4 - Loyalty & Promotion Impact

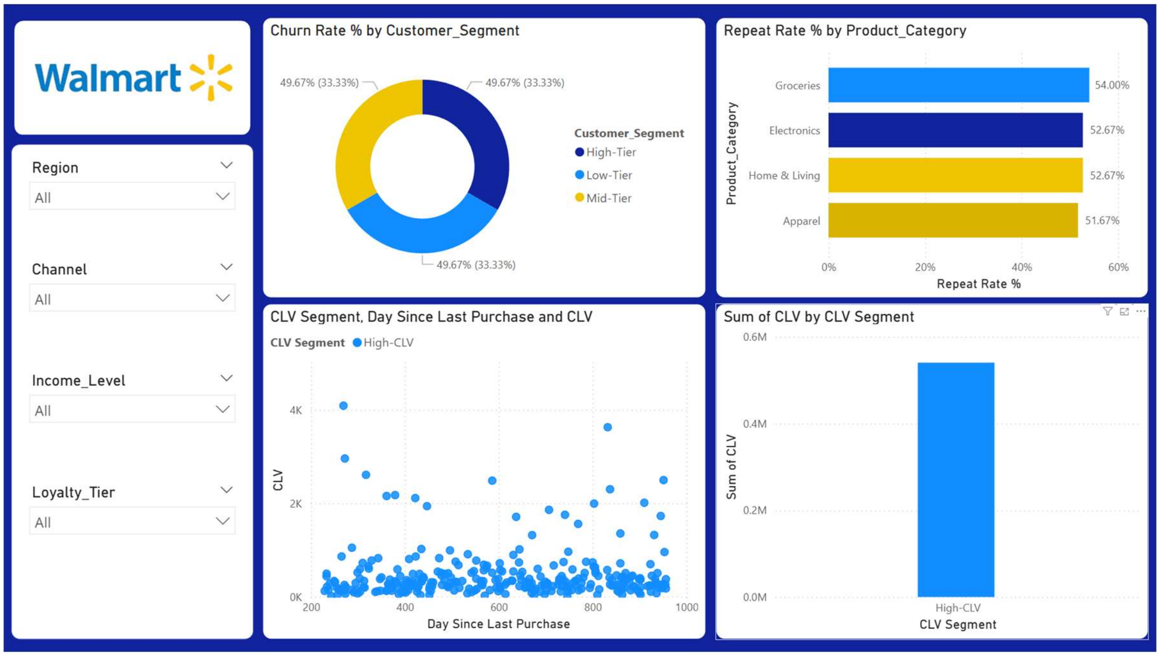
Promotion transaction percentage, average purchase amount comparison, churn rate by loyalty tier, and points earned versus redeemed were analyzed.

7. Task 5 - Store & Channel Insights

Average transaction amount by store type, churn rate by store type, and opening year versus retention rate correlation were visualized.

8. Task 6 – Customer Segmentation

Customer Lifetime Value was calculated and customers were segmented into High and Low CLV. CLV was compared against customer inactivity, loyalty tier, and region.



9. Task 7 - Final Dashboard & Executive Summary

A four-page dashboard was created: KPI Overview, Loyalty & Promotion Impact, Store & Channel Insights, and Customer Segmentation.

10. Recommendations

1. Walmart should focus more on high CLV customers who have become inactive, because losing these customers would result in the highest revenue loss.

2. Target inactive and low-tier customers with personalized promotions to improve retention.

3. Walmart should strengthen loyalty program engagement by making reward redemption easier.

4. Improve underperforming channels with focused customer experience strategies.

11. Conclusion

This project demonstrates how Power BI can convert retail customer data into meaningful insights and actionable business recommendations for improving retention, loyalty participation, and long-term profitability.

Video Link: https://www.loom.com/share/b10928e95c3644b2a7e4c7b34bea3b1f

Power BI Project Link:
https://drive.google.com/drive/folders/1WdUncWjeD8sAPK1itGY23zIWLTIK4Irk?usp=sharing